

Class, Object, Methods

20 June 2026 11:26

Till now we have written most of the code inside :

```
class ConBreak
{
    public static void main(String[] args)
```

We have learned:

1. Variables
2. Operators
3. Conditions
4. Loops

But after doing so many program till now we learned how to organizes data and how java behaves .

What is class ?

Class defines what can object should have and what it should do.

What is object ?

Object is the real instance created from class

What is method ?

Method represents an action or behaviour

What does a Student have ?

What can student do ?

Student has:

1. Name
2. Age
3. Roll num
4. Course

Student can:

1. Study
2. Attend class
3. take exam
4. Notes

Let's connect this with java

Data becomes fields

Action become methods

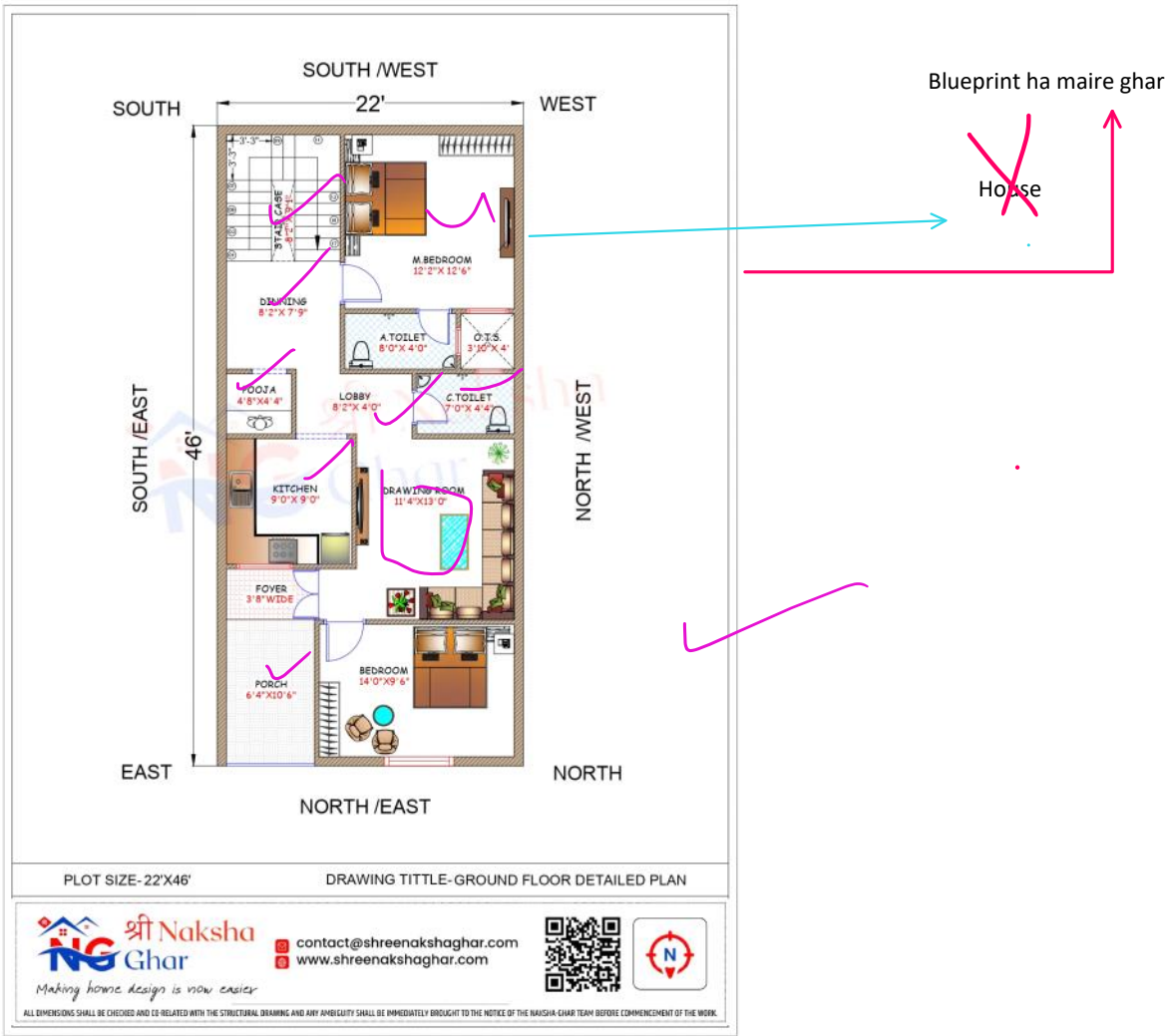
Both are placed inside class

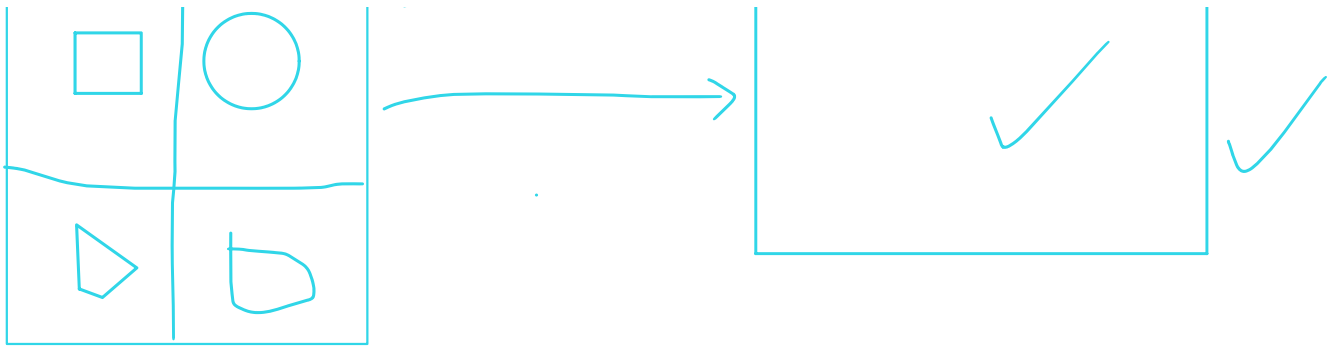
What is class ?

A class is blueprint or template used to create objects

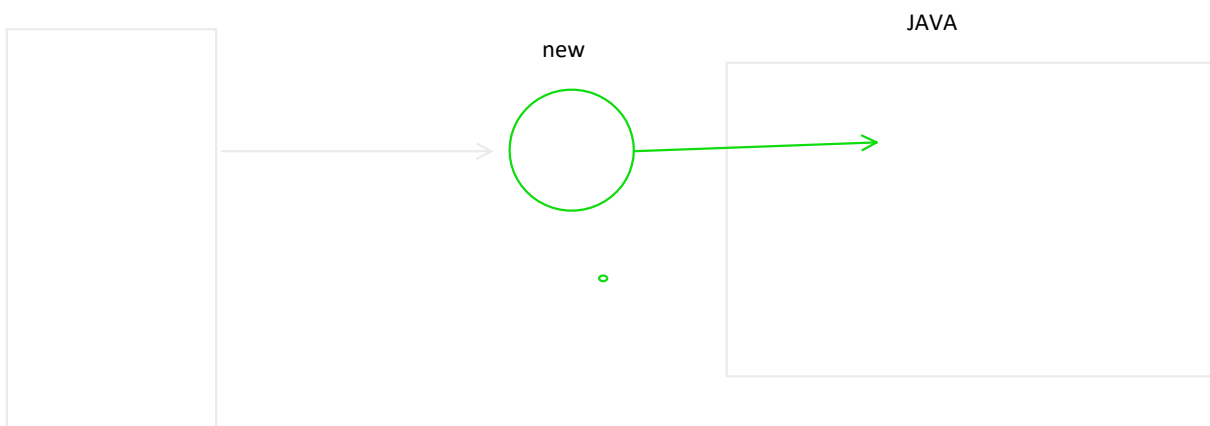
Simple definition:

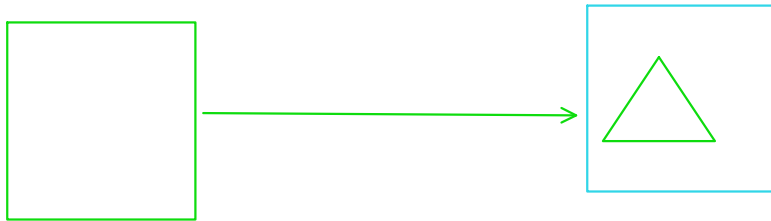
Class = blueprint that defined data and behaviours





BankAccount
EmployeeDetails
ProductDetails





Data Type	Default Value (for fields)
byte	0
short	0
int	0 ✓
long	0L
float	0.0f
double	0.0d
char	'\u0000'
String (or any object)	null ✓
boolean	false

Reference variable:

Student s= new Student();

The variable used to refer to an object is called a reference variable.

S is here reference variable

New Student() create the object

Dot operator

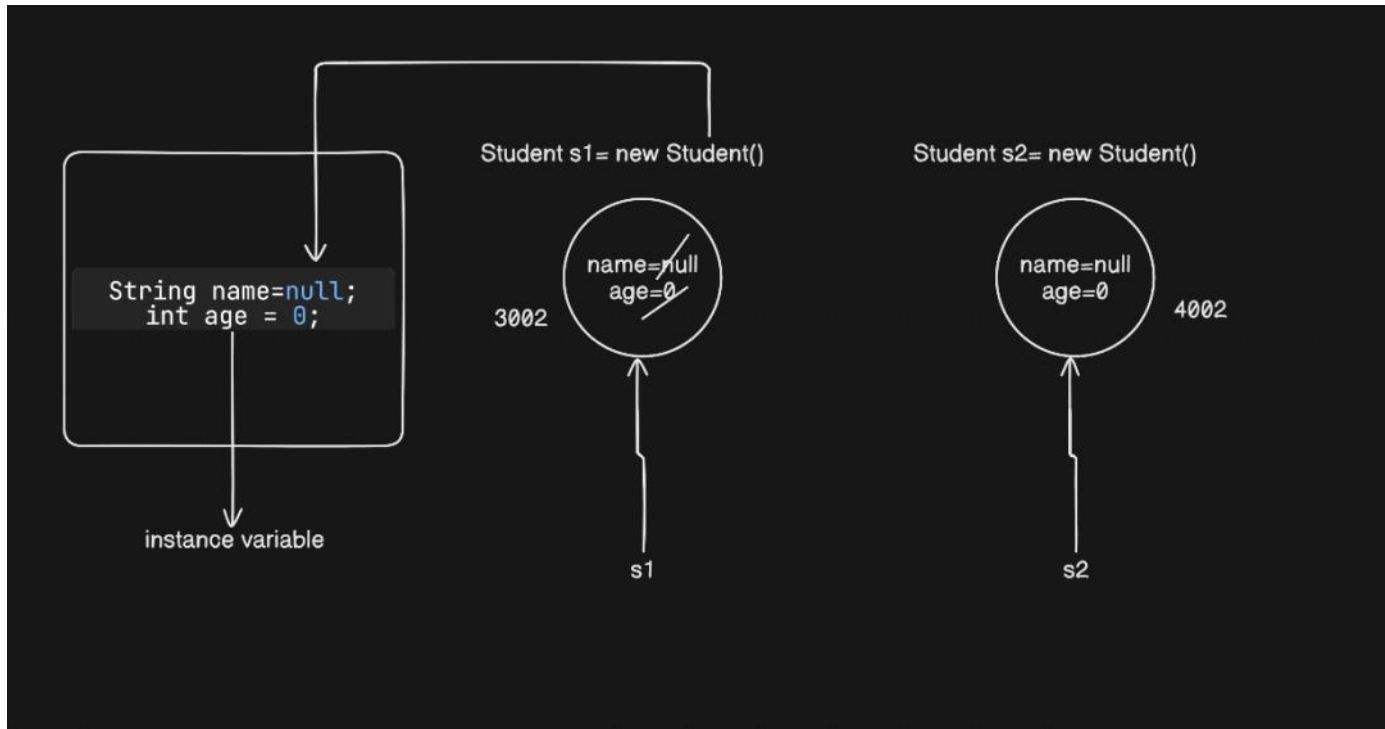
The dot operator is used to access fields and method through reference

Eg:

s.name

s.age

s.study()



What is Method ?

A method is a named block of code that perform a specific task.

Method = reusable block of code that perform an action

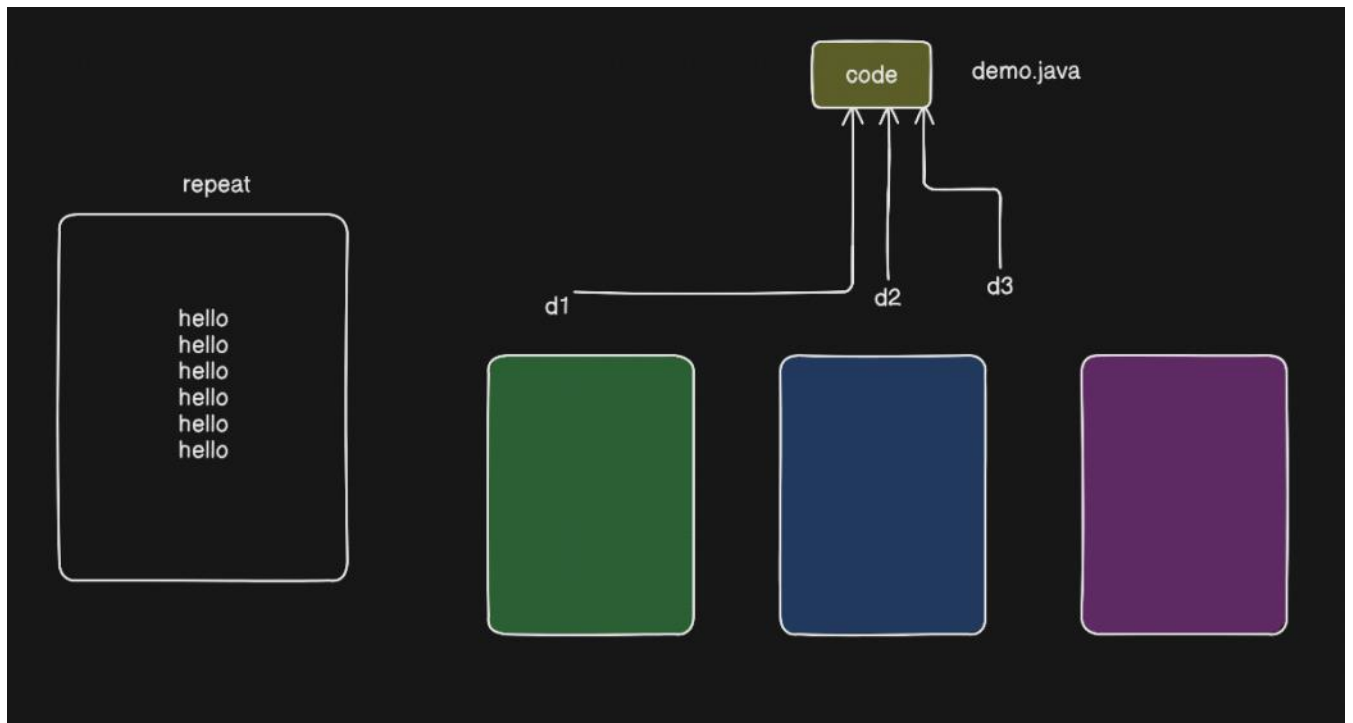
Why we need methods ?

Without method :

```
System.out.println("Name:rahul");  
System.out.println("Age:30");  
System.out.println("Course: Spark 6.0");
```

If we need this logic in many places then how will do it ?

We can use method and use it by reference by variable



Method declaration

```
void study()
{
    System.out.println("Student is studying.....");
}
```

Void : is the return type
 Study : method name
 () contains parameters if any
 {} contains method body

Method call

s.study()

Parameter and Argument

A parameter is a variable declared in method definition.

An argument is the actual value passed during the method call

```
public class Babu {

    void sorry(String name) 1 usage
    {
        System.out.println(name + " sorry bo!o....");
    }
}
```

```

        System.out.println(name + " sorry bolo....");
    }

    public static void main(String[] args) {
        Babu b= new Babu();
        b.sorry( name: "Hariya");
    }
}

```

Method return type

We have learned primitive and non-primitive data types all of these we can use as method return type.

Four common method types

1. No parameters, No return value
2. Parameter, No return value
3. No parameter, return value
4. Parameter, return value

Local variable:

A variable declared inside a method or block is called local variable

```

void run() no usages
{
    int speed=200;
    String brand="BMW";

    System.out.println(speed);
}

```

Note: local variable exist only inside method or block

Instance variable	Local variable
Declare inside class, outside method	Declare inside method or block
Belongs to an object	Belongs to method
Get default values	Must be initialized before use

Accessible from anywhere in class	Access inside only same method
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